

Venkatesh P

AI / ML ENGINEER · RAG · LLM AGENTS · MLOPS

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🌐 [linkedin.com/in/venkatesh-ml](https://www.linkedin.com/in/venkatesh-ml) 🏠 github.com/venkatesh-hyper 🌐 Portfolio

Applied AI/ML Engineer with **end-to-end ownership** of RAG pipelines, LLM agents, and production ML systems. Built a semantic search engine over **42,549 papers at 2,979 QPS** ($p50 = 0.31$ ms, $p95 = 0.53$ ms) and a medical NLP/RAG system deployed on GCP. Hands-on across the full stack: data ingestion → vector retrieval → LLM serving → production deployment. Stack: FastAPI · FAISS · LangChain · Docker · GCP · MLflow · HuggingFace.

Technical Skills

AI Agents & RAG	LangChain, LangGraph, LlamaIndex, ReAct, Multi-Agent, RAGAS	LLMs GenAI	HuggingFace, BERT/GPT/LLaMA, Fine-Tuning (LoRA/QLoRA), Groq API
Vector Search	FAISS (IVFFlat/HNSW/PQ), ChromaDB, Weaviate, Hybrid Search, Reranking	ML & Deep Learning	PyTorch, TensorFlow, Scikit-learn, CNNs, RNNs, Ensemble, XGBoost
Cloud MLOps	GCP (Vertex AI, GCS), AWS (EC2, S3), Docker, MLflow, DVC, GitHub Actions	Data Backend	FastAPI, PySpark, Pandas, SQL (Advanced), PostgreSQL, Redis

Experience

AI/ML Engineer Intern · NASO Technologies

Jan 2024 – Apr 2024

- Designed and shipped **Well.AI** end-to-end: data ingestion → HuggingFace embeddings → ChromaDB → LLM clinical QA, served via **FastAPI on GCP** with Docker Compose and Supabase session logging.
- Engineered prompt workflows for **clinical health reasoning**; structured retrieval context windows to improve response faithfulness and reduce hallucination rate in medical QA tasks.
- Translated business requirements into ML solutions with domain stakeholders; delivered across dev, staging, and prod with a **governance-ready deployment** pipeline.

Machine Learning Researcher · RUSA Project, University of Madras

Apr 2024 – Jul 2024

- Built automated ML pipeline on a **high-dimensional tabular dataset (50K+ features)**: feature engineering → ensemble training (XGBoost + Random Forest) → hyperparameter tuning, achieving **87.6% classification accuracy**.
- Deployed interpretable **Streamlit dashboard** for non-technical stakeholders; implemented **DVC** for full data versioning and experiment tracking on GCP.

Projects

PaperLens — Semantic Search Engine with Agentic RAG

[GitHub](#) · [Live Demo](#)

sentence-transformers · FAISS IVFFlat · FastAPI · Groq LLaMA-3 · ChromaDB · Docker · Streamlit

- Built full retrieval pipeline over **42,549 arXiv papers (20 domains)**: ingestion → 384-dim embeddings → FAISS IVFFlat (62.3 MB index) → RAG summarisation; benchmarked at $p50 = 0.31$ ms, $p95 = 0.53$ ms, **2,979 QPS** — production-grade throughput on a low-resource deployment.
- FastAPI backend with **LRU embedding cache** cut repeated query latency 1,100 ms → **0.94 ms (55× speedup)**; Groq LLaMA-3.3-70b generates structured 4-part paper summaries in ~1,200 ms.

Enterprise Ticket Analysis Bot — Agentic RAG with Tool Use

[GitHub](#)

FastAPI · ChromaDB · Groq Llama-4 Scout · SentenceTransformers · LangChain · Streamlit

- Production RAG pipeline over **70 MB+ enterprise ticket logs**: PDF parsing → chunking → SentenceTransformer embeddings → LangChain ReAct agent with tool use, achieving **p95 retrieval under 200 ms**.
- Implemented **agent memory, planning, and multi-step execution loops**; natural-language querying over enterprise data — applicable to support automation and operational analytics.

Brain Tumor Classifier — Automated Medical Imaging Pipeline

[GitHub](#)

TensorFlow · CNN · Transfer Learning · Data Augmentation · Docker

- Automated preprocessing → augmentation → CNN training → evaluation; achieved **93% accuracy** on 4-class MRI brain tumor classification with transfer learning; containerised inference via Docker.

Education

Master of Computer Applications (MCA)

University of Madras, Chennai 2023–2025 GPA: 7.38/10

Specialisation: Machine Learning, Deep Learning, Data Science

B.Sc. Biotechnology

RKM Vivekananda College, Chennai 2020–2023 GPA: 7.2/10

Foundation in statistics, computational methods, and biological data analysis

Achievements, Leadership & Certifications

- Mentor & AI Educator** — Guided 70+ HSC students (3 yrs) on AI fundamentals and career paths
- Vice President**, Rotaract Club of RKMVC — led 5+ AI awareness events across Chennai
- 2nd Place** — AI Front-End Hackathon 2025, MCC Chennai (25+ colleges)
- Published technical articles on EDA, NLP pipelines, and ML on **Medium**
- AWS Data Engineering Bootcamp** (Udemy, 2024) · Deep Learning Specialization (Coursera, In Progress)